

2D Variation Modeling and Probabilistic Sampling in Long-Term Time Series Forecasting

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Abstract—Long-term time series forecasting (LTTF) is essential for modern industrial infrastructure management and financial decision-making, however, capturing complex temporal dependencies across non-stationary signals remains challenging. In this work, we present an empirical benchmark evaluating TimesNet, a temporal 2D-variation architecture utilizing 2D Fast Fourier Transforms, against DLinear, 1D-ConvNet, and an instance-normalized LSTM across two distinct datasets: hourly electricity consumption and daily exchange rates. Our comparative evaluation reveals that TimesNet significantly outperforms baseline architectures on strongly periodic engineering data by capturing multi-scale intra- and inter-period variations. Conversely, lightweight linear decomposition (DLinear) outperforms complex deep networks on low signal-to-noise financial market series, demonstrating superior robustness against overfitting. Furthermore, we investigate probabilistic forecasting strategies, demonstrating that test-time Monte Carlo (MC) Dropout outperforms parametric Gaussian formulations by preserving sharp peak predictions while generating reliable epistemic uncertainty bounds without architectural modifications. These findings provide clear practical design guidelines, helping engineers select the appropriate model complexity depending on their domain signal-to-noise dynamics in real-world scenarios.

Index Terms—Time Series Forecasting, TimesNet, DLinear, Reversible Instance Normalization, Probabilistic Forecasting, Monte Carlo Dropout.

I. INTRODUCTION

Reliable long-term time series forecasting (LTTF) is crucial for energy grid balancing, resource allocation, and financial decision-making. However, modeling real-world multivariate signals remains challenging due to non-stationary dynamics, direction trends, and overlapping periodic patterns. While deep learning has advanced LTTF, traditional architectures exhibit structural pitfalls: recurrent models (such as RNNs or LSTMs) suffer from error accumulation and vanishing gradients, 1D Temporal Convolutional Networks (like TCNs) are constrained by local receptive fields. Recently, Wu *et al.* [1] introduced TimesNet, which leverages 1D-to-2D Fast Fourier Transforms to capture intraperiod and interperiod variations. Concurrently, Zeng *et al.* [2] demonstrated that simple linear decomposition (DLinear) can outperform complex backbones. In all these studies, existing benchmarks focus predominantly on deterministic point forecasts, leaving a gap in understanding how 2D multi-periodic representations generalize under probabilistic uncertainty across different signal regimes.

In this project report, we perform an independent empirical evaluation of TimesNet against DLinear, 1D-ConvNet, and an instance-normalized LSTM baseline across periodic engineering data (*Electricity*) and non-stationary financial data

(*Exchange Rate*). Furthermore, to address the lack of uncertainty quantification in critical deployments or risky settings, we extend TimesNet to probabilistic forecasting. We compare a Parametric Gaussian formulation against test-time Monte Carlo (MC) Dropout, proving that MC Dropout yields well calibrated epistemic uncertainty bounds without smoothing out sharp temporal spikes.

Contributions: Our main contributions are four: **(i) Probabilistic Extension:** We formulate and contrast Parametric Gaussian NLL against MC Dropout ($N = 50$), showing that MC Dropout preserves peak accuracy while providing reliable confidence bounds; **(ii) Independent Benchmark:** We evaluate TimesNet and DLinear across high-signal periodic and low signal-to-noise financial regimes, providing model selection guidelines; **(iii) Ablation Studies:** We evaluate Reversible Instance Normalization (RevIN), execute FFT top- k frequency sweeps ($k \in \{1, 3, 5, 7\}$), and measure horizon error drift up to very long terms ($H = 720$).

The remainder of this paper is organized as follows: Section II reviews related work. Sections III and IV detail the data pipeline and describe benchmark temporal signals. Section V presents the learning framework and probabilistic extensions, Section VI evaluates performance, and Section VII concludes the paper.

II. RELATED WORK

Early deep learning approaches for long-term time series forecasting (LTTF) relied on Recurrent Neural Networks (LSTMs) [3] and 1D Temporal Convolutional Networks (TCNs) [4]. However, recurrent models suffer from error accumulation over extended autoregressive steps, while 1D-TCNs are constrained by local receptive fields [1]. To model global dependencies, more complex Transformer architectures like Autoformer [5] and FEDformer [6] introduced specialized self-attention and seasonal trend decomposition. However, attention mechanisms often struggle to extract reliable temporal relationships from noisy time steps [1].

Unlike Transformer complexity, Zeng *et al.* [2] introduced DLinear, demonstrating that a single linear layer applied to trend and seasonal components outperforms complex attention backbones. However, DLinear has a fixed temporal mapping that lacks adaptive frequency extraction, so it becomes sub-optimal for complex multi-periodic signals [1]. To overcome this, Wu *et al.* [1] proposed TimesNet, leveraging 1D-to-2D Fast Fourier Transforms (FFT) to transform 1D sequences into 2D tensors, capturing intraperiod and interperiod variations via

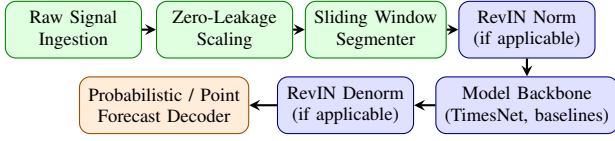


Fig. 1: High-level block diagram showing the 6-stage pipeline.

2D Inception convolutions. Additionally, Kim *et al.* [7] introduced Reversible Instance Normalization (RevIN) to eliminate window distribution shifts in non-stationary signals.

While these models focus primarily on deterministic point forecasts, some critical applications (i.e. energy deployment or financial decisions) require probabilistic risk quantification. Frameworks like DeepAR [8] model parametric likelihood distributions, whereas non-parametric Bayesian approximations like test-time Monte Carlo (MC) Dropout [9] capture epistemic uncertainty without architectural modifications. In this work, we extend TimesNet with parametric Gaussian heads and non-parametric Monte Carlo (MC) Dropout, evaluating their uncertainty calibration across contrasting data modalities.

III. PROCESSING PIPELINE

The system architecture (Fig. 1) comprises six interconnected operational modules in three phases:

- 1) **Data Ingestion & Standardization:** Loads raw temporal archives and converts them into continuous multi-variate matrices. Non anticipative channel statistics are calculated strictly on training splits to standardize all partitions.
- 2) **Sliding Window Segmentation:** Extracts uniform historical lookback contexts (L) and target prediction horizons (H) using a Sliding Window DataLoader mechanism.
- 3) **Adaptive Instance Normalization (RevIN Norm):** Applied only to models wrapped in RevIN (TimesNet, LSTM), standardizes the mean and variance of each sample, mitigating the shifts in the input distribution. Linear (DLinear) and 1D-convolutional baselines process lookback windows directly.
- 4) **Neural Network Backbone:** Processes lookback tensors with the selected model, executing 2D FFT period discovery in TimesNet, moving-average seasonal-trend decomposition in DLinear, or spatial/temporal feature extraction in 1D-ConvNet and LSTM.
- 5) **Instance Denormalization (RevIN Denorm):** Restores projected feature outputs back to the original data scale using saved statistics for each window, only for RevIN-wrapped models.
- 6) **Output Decoding & Uncertainty Module:** Projects feature representations to horizon H thanks to linear mapping. Deterministic models output point forecasts, while probabilistic extensions estimate dual Gaussian parameters (μ, σ^2) or execute test-time Monte Carlo (MC) Dropout sampling (drawing $N = 50$ samples) to output point estimates with 90% confidence interval.

IV. SIGNALS AND FEATURES

A. Measurement Setup and Signal Formatting

We tested our models on two different real-world data:

Electricity: Energy demand sampled hourly (1h) from $C = 321$ industrial clients over $T = 26,304$ steps (8,443,584 total points). As shown in Fig. 2, it represents a high-signal engineering domain with dominant 24h daily periodicity [10].

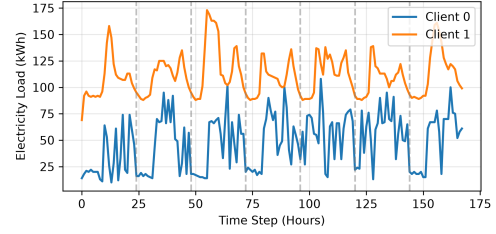


Fig. 2: Electricity dataset: Zoomed-in 7-day view displaying daily cycles.

Foreign Exchange Rate: Sampled daily (1d) between $C = 8$ currency pairs in $T = 7,588$ steps (60,704 total points). As illustrated in Fig. 3, it captures a low signal-to-noise financial domain governed by non-stationary, near to random walk dynamics [10].

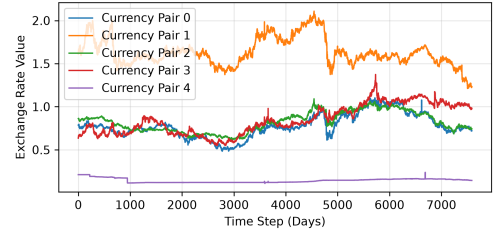


Fig. 3: Exchange Rate dataset: Full timeline ($\approx 7,500$ d).

B. Signal Pre-processing and Partitioning

To eliminate scale disparities without introducing data leakage, channel statistics (μ_c, σ_c) are calculated exclusively on the training split:

$$\mu_c = \frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} x_{t,c}, \quad \sigma_c = \sqrt{\frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} (x_{t,c} - \mu_c)^2 + \epsilon}, \quad (1)$$

where $x_{t,c}$ is the value at step t for channel c . The stored parameters are used to normalize the validation and test sets. According to the original paper, the datasets are chronologically divided into 70% training, 10% validation, and 20% testing splits to preserve temporal ordering. Table 1 summarizes the attributes of the data set.

TABLE 1: Dataset Structural Metrics

Metric	Electricity	Exchange Rate
Total Time Steps (T)	26,304	7,588
Channels / Variates (C)	321	8
Total Data Points	8,443,584	60,704
Sampling Frequency	Hourly (1 h)	Daily (1 d)
Global Mean (μ)	2,538.7916	0.6947
Global Std Dev (σ)	15,027.5702	0.4761

C. Feature Extraction and Sliding Window Strategy

Pre-processed series are segmented into temporal sliding windows. For index i , lookback sequence $\mathbf{X}_i \in \mathbb{R}^{L \times C}$ and target sequence $\mathbf{Y}_i \in \mathbb{R}^{(O+H) \times C}$ are extracted as:

$$\mathbf{X}_i = \{\mathbf{x}_t\}_{t=i}^{i+L-1}, \quad \mathbf{Y}_i = \{\mathbf{x}_t\}_{t=i+L}^{i+L+H-1}, \quad (2)$$

where lookback length $L = 96$ and forecast horizon $H = 96$. NN backbones extract feature representations directly from these formatted window tensors.

V. LEARNING FRAMEWORK

A. Distribution Shift Mitigation (RevIN)

To combat non-stationarity between lookbacks and targets, TimesNet and LSTM incorporate Reversible Instance Normalization (RevIN). For input $\mathbf{X} \in \mathbb{R}^{B \times L \times C}$, RevIN computes instance mean $\mu_{\mathbf{X}}$ and variance $\sigma_{\mathbf{X}}^2$ along temporal dimension L :

$$\mu_{\mathbf{X}} = \frac{1}{L} \sum_{t=1}^L \mathbf{x}_{t,:}, \quad \sigma_{\mathbf{X}} = \sqrt{\frac{1}{L} \sum_{t=1}^L (\mathbf{x}_{t,:} - \mu_{\mathbf{X}})^2 + \epsilon}, \quad (3)$$

where $\epsilon = 10^{-5}$. Affine parameters $\gamma, \beta \in \mathbb{R}^{1 \times C}$ scale normalization and inverse denormalization:

$$\hat{\mathbf{X}} = \gamma \odot \left(\frac{\mathbf{X} - \mu_{\mathbf{X}}}{\sqrt{\sigma_{\mathbf{X}}^2 + \epsilon}} \right) + \beta, \quad \hat{\mathbf{Y}} = \left(\frac{\mathbf{Y}_{\text{norm}} - \beta}{\gamma} \right) \odot \sqrt{\sigma_{\mathbf{X}}^2 + \epsilon} + \mu_{\mathbf{X}}. \quad (4)$$

B. Baseline Architectures

We evaluate three comparative baseline NN backbones:

- **Decomposed Linear Model (DLinear):** Decomposes the input $\mathbf{X}$ into a trend component $\mathbf{X}_{\text{Trend}}$ with a moving-average filter ($k = 25$) and a seasonal component $\mathbf{X}_{\text{Seas}} = \mathbf{X} - \mathbf{X}_{\text{Trend}}$:

$$\mathbf{X}_{\text{Trend}} = \text{AvgPool1D}(\text{Pad}(\mathbf{X})). \quad (5)$$

Linear weights $\mathbf{W}_S, \mathbf{W}_T \in \mathbb{R}^{H \times L}$ project temporal components from L to H :

$$\hat{\mathbf{Y}} = (\mathbf{X}_{\text{Seas}} \mathbf{W}_S^\top + \mathbf{X}_{\text{Trend}} \mathbf{W}_T^\top)^\top \in \mathbb{R}^{B \times H \times C}. \quad (6)$$

- **Stacked Depthwise 1D-ConvNet:** Combines $N = 2$ residual blocks applying depthwise temporal convolution (groups = $C, k = 3$), GELU activation, dropout ($p = 0.1$), and 1×1 pointwise projection:

$$\mathbf{H}^{(l)} = \text{Conv1D}_{1 \times 1} \left(\text{Dropout} \left(\sigma \left(\text{DepthwiseConv1D} \left(\mathbf{H}^{(l-1)} \right) \right) \right) \right) + \mathbf{H}^{(l-1)}, \quad (7)$$

where $\mathbf{H}^{(0)} = \mathbf{X}^\top$. The shared linear layers project the output temporal steps from $L \rightarrow H$.

- **Channel independent LSTM:** Captures the dynamics of the recurrent state $\mathbf{h}_t = \text{LSTM}(\hat{\mathbf{x}}_t, \mathbf{h}_{t-1})$ wrapped with RevIN. Hidden states are projected through a shared linear layer mapping $(L \cdot d_h) \rightarrow H$.

C. Primary Architecture: TimesNet

TimesNet maps 1D sequences into 2D intraperiod and interperiod tensors across $N_e = 2$ stacked TimesBlock modules.

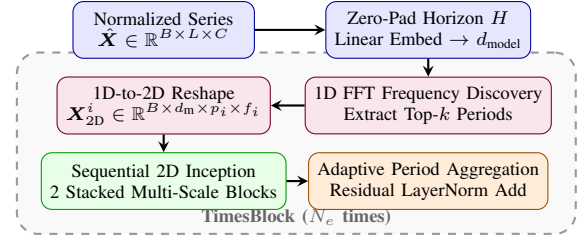


Fig. 4: Detailed schematic of a residual TimesBlock.

1) *FFT Period Discovery & 2D Reshaping:* Target zero padding placeholder $\mathbf{0} \in \mathbb{R}^{B \times H \times C}$ is appended to lookback $\hat{\mathbf{X}}$ to form a sequence of length $T = L + H$, projected to latent dimension d_{model} . With Fast Fourier Transform (FFT) it identifies top- k dominant frequencies along dimension T :

$$\mathbf{A} = |\text{FFT}(\mathbf{E})|, \quad \mathbf{a} = \frac{1}{B \cdot d_{\text{model}}} \sum_{b=1}^B \sum_{d=1}^{d_{\text{model}}} \mathbf{A}_{b,:,d}. \quad (8)$$

Peak frequency indices are related to period lengths $p_i = \lceil T/f_i \rceil$. For each period p_i , zero-padded 1D tensors are reshaped into 2D tensors $\mathbf{X}_{2D}^i \in \mathbb{R}^{B \times d_{\text{model}} \times p_i \times (L_i/p_i)}$.

2) *2D Inception Convolutions & Adaptive Period Aggregation:* Each 2D tensor $\mathbf{X}_{2D}^i$ is processed by parallel 2D Inception kernels ($1 \times 1, 3 \times 3, 5 \times 5, \dots$):

$$\mathcal{F}_{2D}(\mathbf{X}_{2D}^i) = \text{Inception2D}(\text{GELU}(\text{Inception2D}(\mathbf{X}_{2D}^i))). \quad (9)$$

Output 2D representations are reshaped to 1D, truncated to length T , and aggregated across k periods using softmax amplitude weights w_i :

$$w_i = \frac{\exp(a_{f_i})}{\sum_{j=1}^k \exp(a_{f_j})}, \quad \mathbf{E}_{\text{out}} = \text{LayerNorm} \left(\sum_{i=1}^k w_i \hat{\mathbf{X}}_{1D}^i + \mathbf{E} \right). \quad (10)$$

Linear projection $\text{Linear}_{(d_{\text{model}} \rightarrow C)}$ gives target predictions $\hat{\mathbf{Y}}_{\text{norm}} = \mathbf{E}_{\text{out}}[:, -H :, :]$.

D. Probabilistic Uncertainty Extensions

We add two non-deterministic extensions into TimesNet architecture:

Parametric Gaussian Head (ProbabilisticTimesNet): Uses TimesNet feature representations $\mathbf{H}_{\text{base}} = \text{TimesNet}(\mathbf{X})$ as input of two linear heads estimating mean and standard deviation:

$$\mu_{\Theta} = \text{head}_{\mu}(\mathbf{H}_{\text{base}}), \quad \sigma_{\Theta} = \text{Softplus}(\text{head}_{\sigma}(\mathbf{H}_{\text{base}})) + 10^{-3}. \quad (11)$$

Quantile bounds construct 90% and 99% confidence intervals:

$$\text{CI}_{90\%} = \mu_{\Theta} \pm 1.645 \cdot \sigma_{\Theta}, \quad \text{CI}_{99\%} = \mu_{\Theta} \pm 2.576 \cdot \sigma_{\Theta}. \quad (12)$$

Monte Carlo (MC) test-time Dropout: Evaluation executes $N = 50$ stochastic forward passes with activated dropout with higher rate ($p = 0.3$), from obtained samples we extract trajectory mean μ_{MC} and empirical 90% percentile bounds:

$$\mu_{\text{MC}} = \frac{1}{N} \sum_{n=1}^N \hat{\mathbf{Y}}^{(n)}, \quad [\mathbf{y}_{0.05}, \mathbf{y}_{0.95}] = \text{Percentile}_{5,95} \left(\{\hat{\mathbf{Y}}^{(n)}\}_{n=1}^N \right). \quad (13)$$

E. Optimization Strategy and Training Conditions

Deterministic models minimize Mean Squared Error ($\mathcal{L}_{\text{MSE}}$) with gradient norm clipping (≤ 5.0). ProbabilisticTimesNet minimizes Gaussian Negative Log-Likelihood (NLL) regularized with weighted MSE ($\lambda = 2.0$) to avoid smoothing:

$$\mathcal{L}_{\text{Prob}} = \frac{1}{BHC} \sum_{b=1}^B \sum_{t=1}^H \sum_{c=1}^C \left(\frac{\log \sigma_{b,t,c}^2}{2} + \frac{(y_{b,t,c} - \mu_{b,t,c})^2}{2\sigma_{b,t,c}^2} \right) + 2\mathcal{L}_{\text{MSE}}(\mu, \mathbf{Y}) \quad (14)$$

Optimization utilizes Adam ($\eta = 10^{-3}$, weight decay 10^{-4}) with ‘ReduceLROnPlateau’ decay ($\gamma = 0.5$, patience 2). Configurations are detailed in Table 2.

TABLE 2: Hyperparameter and Architectural Specifications

Configuration Parameter	Electricity	Exchange Rate
Lookback Window (L)	96	96
Forecast Horizon (H)	96	96
Batch Size (B)	32	32
Training Epochs	15	15
Initial Learning Rate (η)	10^{-3}	10^{-3}
Weight Decay	10^{-4}	10^{-4}
Model Capacity (d_{model})	64	36
Inception Kernels (N_k)	4	3
Top- k Fourier Periods	5	5
Stacked Blocks (N_e)	2	2
Reversible IN (RevIN)	Enabled	Enabled
LSTM Hidden Dimension (d_h)	128	32
DLinear Moving Average (k)	25	25
1D-ConvNet Kernel Size (k)	3	3

Model experiments train backbones for 7 epochs across:

- 1) **Horizon Generalization:** extended horizons $H \in \{96, 192, 336, 720\}$.
- 2) **Top- k Period Sensitivity:** Fourier period counts $k \in \{1, 3, 5, 7\}$.
- 3) **RevIN Distribution Shift:** assessing performance with and without instance normalization.

FFT interpretability extracts 1D Real FFT spectral peaks along L , mapping top- k frequencies to physical periods $T_h = L/f_i$. Execution runs locally on PyTorch using Apple Silicon Metal Performance Shaders (MPS) acceleration.

VI. RESULTS

A. Benchmark Deterministic Performance

We evaluated all models DLinear, 1D-ConvNet, LSTM, and TimesNet, across Electricity ($C = 321$) and Exchange Rate ($C = 8$) for the horizon $H = 96$ (Table 3).

On Electricity dataset, TimesNet achieves the best accuracy, because its 2D FFT mapping isolates overlapping daily and weekly cycles. Conversely, on Exchange Rate dataset,

TABLE 3: Quantitative Performance Benchmark ($H = 96$)

Model Architecture	Electricity ($C = 321$)		Exchange Rate ($C = 8$)	
	Test MSE	Test MAE	Test MSE	Test MAE
DLinear	0.1953	0.2788	0.0787	0.1999
1D-ConvNet	0.2112	0.3120	0.0963	0.2388
Instance-Norm. LSTM	0.3599	0.4212	0.1158	0.2414
TimesNet	0.1714	0.2751	0.1452	0.2771

DLinear performs better, because its moving average trend decomposition handles non stationary random walks without fitting the high frequency pseudoperiodic noise that makes TimesNet fail. Across both datasets, LSTM suffers from autoregressive error drift, while depthwise 1D-ConvNet provides steady baseline performance even without global multi periodic capability.

B. Convergence and Visual Prediction Dynamics

Loss trajectories (Fig. 5) show smooth convergence on Electricity, where TimesNet stabilizes by Epoch 2 with the lowest validation loss thanks to RevIN. In contrast, validation curves fluctuate on Exchange Rate, caused by non-stationary noise. Forecast overlays ($H = 96$, Fig. 6) confirm these dynamics: on Electricity, TimesNet accurately tracks sharp periodic demand spikes, while 1D-ConvNet oscillates and LSTM oversmooths. On Exchange Rate, DLinear closely tracks directional trend drift, while TimesNet fits artificial high frequency noise.

C. Model Ablation and Sensitivity Analysis

1) **Horizon Generalization Dynamics:** Performance scaling across horizons $H \in \{96, 192, 336, 720\}$ is reported in Table 4.

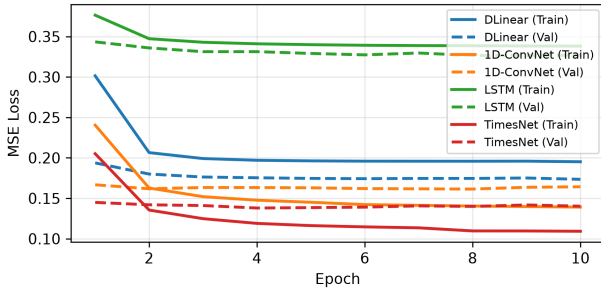
TABLE 4: Horizon Generalization Performance (Test MSE)

Model	Electricity ($C = 321$)			
	$H=96$	$H=192$	$H=336$	$H=720$
TimesNet	0.1751	0.1863	0.3760	0.8827
DLinear	0.1961	0.1961	0.2082	0.2441
1D-ConvNet	0.2112	0.2063	0.2119	0.2542
LSTM	0.3606	0.3753	0.3741	0.4027
Model	Exchange Rate ($C = 8$)			
	$H=96$	$H=192$	$H=336$	$H=720$
TimesNet	0.1548	0.2799	0.4889	1.1631
DLinear	0.0805	0.1666	0.2701	0.7331
1D-ConvNet	0.1179	0.2149	0.4636	0.7913
LSTM	0.1407	0.2117	0.3990	0.9598

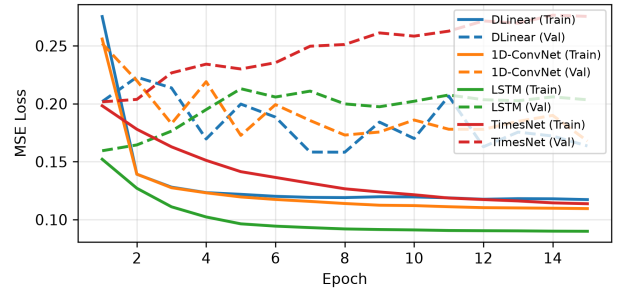
In Electricity, TimesNet leads at moderate horizons ($H \leq 192$), but DLinear is more resilient at $H = 720$ by avoiding 2D convolutional error propagation. In Exchange Rate, DLinear consistently achieves the lowest error across all horizons, confirming that direct linear maps better predict non-stationary trend drift.

2) **Sensitivity to Fourier Period Depth (k):** Table 5 summarizes TimesNet sensitivity to selected period counts ($k \in \{1, 3, 5, 7\}$).

On Electricity, errors stabilize between $k = 3$ and $k = 5$, showing that 3 to 5 dominant frequencies capture the dynamics of true signal without fitting noise. On Exchange Rate, increasing k to 7 reduces Test MSE, as aggregating weaker pseudo-periodic features provides beneficial ensemble smoothing.

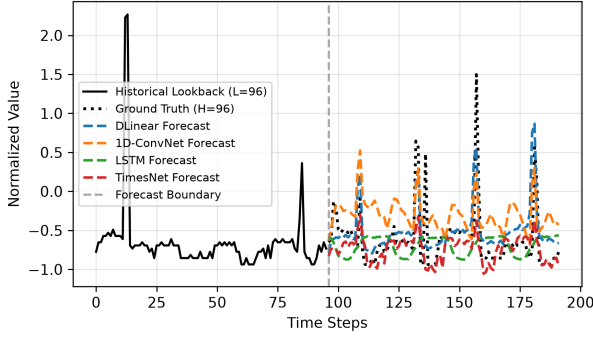


(a) Electricity Loss Curves

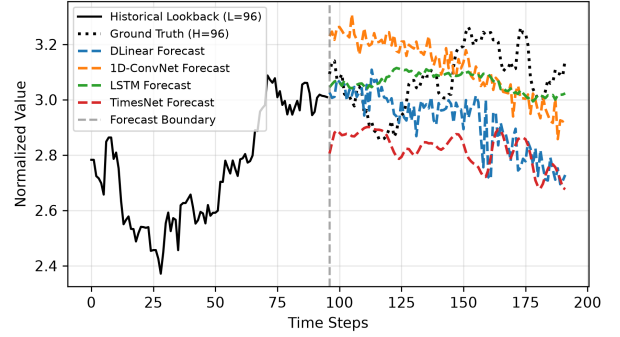


(b) Exchange Rate Loss Curves

Fig. 5: Training and validation loss trajectories across epochs.



(a) Electricity Multi-Peak Tracking



(b) Exchange Rate Trend Tracking

Fig. 6: Visual forecast overlays comparing ground truth against predictions.

TABLE 5: TimesNet Sensitivity to Fourier Period Depth (k)

Setting	Electricity ($C = 321$)		Exchange Rate ($C = 8$)	
	Test MSE	Test MAE	Test MSE	Test MAE
$k = 1$	0.1800	0.2787	0.1578	0.2906
$k = 3$	0.1736	0.2776	0.1595	0.2874
$k = 5$	0.1744	0.2785	0.1516	0.2815
$k = 7$	0.1741	0.2781	0.1423	0.2759

3) *Impact of Instance Normalization (RevIN)*: Disabling RevIN (Table 6) severely degrades Test MSE by 64.67% on Electricity and 95.01% on Exchange Rate, this proves that window-level normalization is essential for 2D convolutional stability.

TABLE 6: Ablation Summary of RevIN

Configuration	Electricity ($C = 321$)		Exchange Rate ($C = 8$)	
	Test MSE	Test MAE	Test MSE	Test MAE
With RevIN	0.1736	0.2782	0.1408	0.2745
Without RevIN	0.4915	0.5169	2.8213	1.3958

D. Domain Interpretability via FFT Analysis

To inspect learned physical representations, Fig. 7 visualizes Fast Fourier Transform (FFT) amplitude spectra ($L = 96$) and top-5 selected period peaks.

On Electricity (Fig. 7a), the peak spectral amplitude occurs at frequency index $f = 4$ ($T = 24.0h$), followed by harmonic peaks at $f = 8$ (12.0h), $f = 12$ (8.0h), and $f = 16$ (6.0h). This proves TimesNet automatically isolates

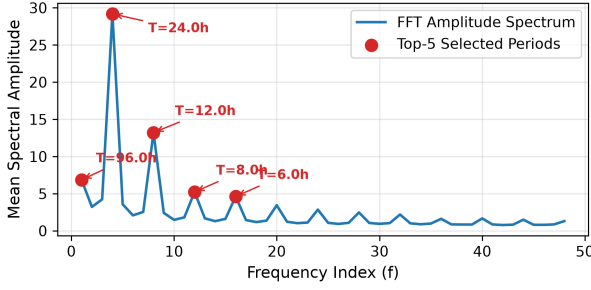
true real-world power consumption cycles. On Exchange Rate (Fig. 7b), the spectrum shows a power-law decay centered at low frequencies, confirming that TimesNet extracts broad low-frequency trend components typical of noise rather than intra-day harmonics of non-periodic market data.

E. Non-Deterministic Probabilistic Forecasting

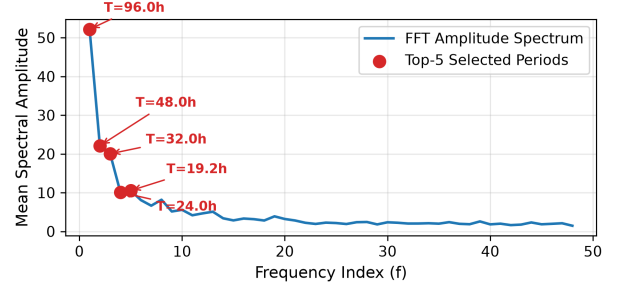
We evaluate two probabilistic extensions for TimesNet: ProbabilisticTimesNet (Parametric Gaussian Likelihood) and Monte Carlo (MC) test-time Dropout (with rate $p = 0.3$, $N = 50$ samples). Visual predictions and uncertainty bounds are summarized in Fig. 8.

Parametric Gaussian Estimation: On Electricity (Fig. 8a), ProbabilisticTimesNet automatically expands its margin of error during sharp demand spikes and shrinks it when things are steady, successfully capturing extreme peaks. On financial data, it keeps a wide margin of error overall because currency prices are naturally noisy and unpredictable.

Epistemic Uncertainty via MC Dropout: On Electricity (Fig. 8c), MC Dropout produces neat, narrow confidence bands that cleanly track repeating daily patterns. On currency data, however, the band stays too tight. When the real exchange rate suddenly trends upward, it shoots right past the predicted upper bound. This proves MC Dropout struggles to anticipate sudden, unexpected trend changes, leading to underestimating tail risk.

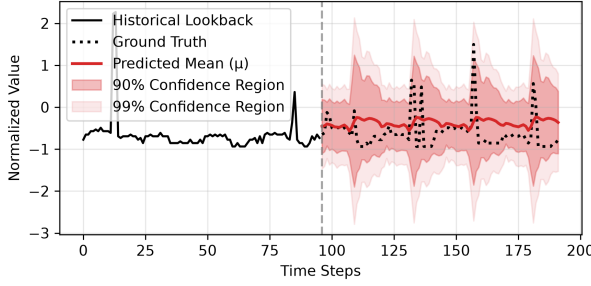


(a) Electricity Spectral Peaks

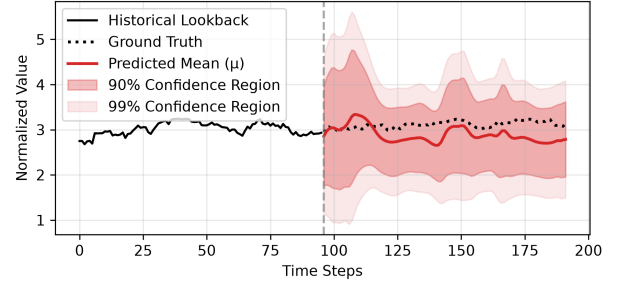


(b) Exchange Rate Spectral Peaks

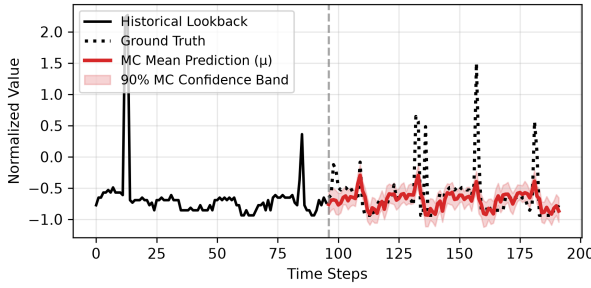
Fig. 7: Fast Fourier Transform (FFT) amplitude spectra ($L = 96$) and top-5 discovered period peaks.



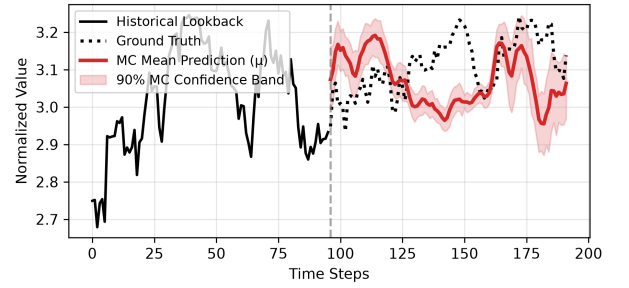
(a) Electricity Gaussian Bounds



(b) Exchange Rate Gaussian Bounds



(c) Electricity MC Dropout Bounds



(d) Exchange Rate MC Dropout Bounds

Fig. 8: Probabilistic forecasts featuring parametric Gaussian bounds (90% and 99% CIs) and non-parametric MC Dropout empirical ribbons (90% CI).

VII. CONCLUDING REMARKS

In this work, we evaluated a probabilistic time-series forecasting framework built on TimesNet, combining 2D Fourier multi-periodicity transformations with parametric Gaussian heads and non-parametric Monte Carlo Dropout sampling. Empirical benchmarks show that while 2D Inception backbones excel at isolating overlapping temporal patterns on periodic series (Electricity), simple linear projections (DLinear) remain superior for trend extrapolation on non-stationary series (Exchange Rate).

These findings establish clear application guidelines: 2D convolutional frequency decomposition should be prioritized for physical systems with stable periodic rhythms, whereas lightweight linear baselines are better for trend-dominated and noisy financial settings. Future extensions could explore

dynamic adaptive frequency filtering to prevent noise fitting.

a) Lessons Learned and Implementation Challenges:

Executing this project provided key insights into empirical deep learning. We learned to embrace contrasting architectural results: instead of viewing TimesNet’s limitations on financial trends as a failure, we accepted these contrasting results as key insights into domain-specific data dynamics. The primary technical challenges came from the ambiguities in the literature regarding the exact placement of RevIN and internal dropout mechanisms, which required the verification of source-code to reach the accuracy and the results on the original paper. Finally, designing novel architectural extensions under real-world computational constraints offered a challenging yet valuable learning experience.

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